

Matilde Dolfato

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EDUCATION

Bocconi University, Milan, Italy

M.Sc. Artificial Intelligence

Sep 2024–Present

- Courses: Linear Algebra, Algorithms for Opt and Inference, Deep Learning, Geometric Computer Vision, Information Theory, PDE, Complex Systems, Optimal Transport, Real and Fourier Analysis (audit)
- GPA: 29.2/30

B.Sc. Economic and Social Sciences

2021–2024

- Advisor: Prof. Jérôme Adda
- Final work: *Can Machine Learning Aid Econometrics?*, grade: 110/110 cum laude

Universitat Pompeu Fabra, Barcelona, Spain

Exchange Program

2024

Università degli Studi di Milano Statale

B.Sc. Biotechnology, Single Courses

2022–2023

EXPERIENCE

Master's Thesis Research

Jul 2026–Present

MaLGa – Machine Learning Genoa Center, University of Genova

- Advisor: Prof. Lorenzo Rosasco
- Research area: learning theory and generalization bounds

Lean Seminar organizer

Mar 2026–Jun 2026

Department of Decision Sciences, Bocconi

- Co-organizing a *reading seminar* on the Lean proof assistant
- Took part in *INdAM Meeting: LftCM 2026* (totally funded)

Research fellow

Nov 2025–Dic 2025

NOVA Information Management School, Lisbon

- Supervisor: Prof. Mauro Castelli
- Research area: explaining *double descent* phenomenon, what underlies stochastic gradient descent preference for smoothness, what is smoothness

Visiting Student

Apr 2025–Nov 2025

Bocconi Institute for Data Science and Analytics (BIDSA)

- Mentor: Prof. Elia Brué
- Research area: Ricci curvature bounds for discrete spaces, optimal transport, convergence of diffusion processes and functional inequalities in spectral theory
- *Technical notes* and two presentations (1,2)

Founding and Board Member

Sep 2025–Present

Bocconi Student Association for Mathematical Sciences

- Gave a *lecture on optimal transport*

Reading Seminars Information Geometry (*Technical notes*)

Spring 2025

Research Internship Algebraic Topology

Spring 2025

Department of Decision Sciences, Bocconi

SELECTED PROJECTS

saLM-1: Self-Aware Language Model Ensemble (LLM ensemble architecture)

Adversarial Robustness Pipeline (CNN adversarial training)

Wordle Guesser (Entropy-based guesser)

Movie Recommendation System (Web app)

SKILLS

Programming: Lean, Python (proficient), R, STATA (advanced), C, C++ (beginner)

Languages: Italian (native), English (proficient), Spanish (advanced)

OTHER INTERESTS

Sailing, surfing, windsurfing; volleyball (ex-agonistica); mountain hiking and skiing; piano; theater and cinema